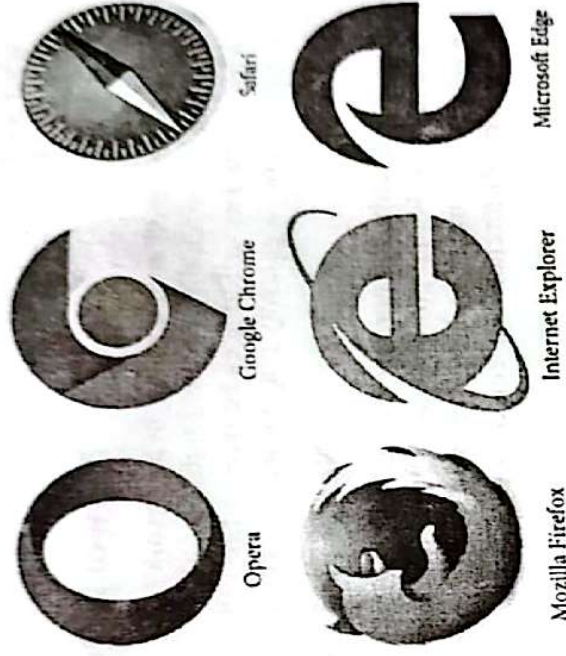


CHAPTER – 8

INTERNET AND WEB

8.1 Web Browser

A web browser is a software application used to access and interact with content on the World Wide Web. It acts as a bridge between the user and the internet, allowing users to retrieve, display, and navigate web pages. Browsers interpret and display HTML code, which is the standard markup language for creating web pages, along with other technologies like CSS (Cascading Style Sheets) for styling and JavaScript for dynamic content. Modern web browsers also support various multimedia formats, enabling the playback of videos, audio, and interactive graphics directly within the browser window.

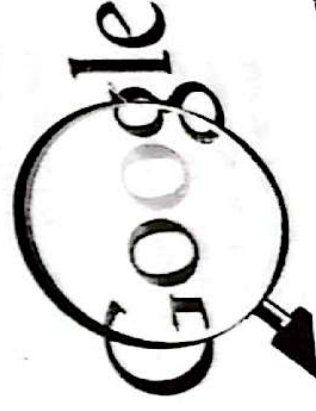


Web browsers typically feature a user-friendly interface that includes an address bar for entering URLs, a search bar, navigation buttons (back, forward, refresh), bookmarks for saving favorite sites, and tabs for managing multiple web pages in a single window. They often come with built-in tools such as download managers, password managers, and privacy controls. The performance of a browser is influenced by its rendering engine, which determines how quickly and accurately it can display web content. Popular web browsers like Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari also offer extensions or add-ons, allowing users to customize their browsing experience with additional functionalities.

Security is a crucial aspect of web browsers. They incorporate features like HTTPS support, sandboxing, and security warnings to protect users from malicious websites, phishing attacks, and unauthorized tracking. Browsers regularly update their security protocols to address vulnerabilities and keep users safe. Additionally, many browsers provide incognito or private browsing modes, which prevent the storage of browsing history, cookies, and other personal data on the user's device. This helps maintain user privacy and is particularly useful when accessing the web from shared or public computers.

8.2 Search Engine

A search engine is a specialized software system designed to help users find information on the internet quickly and efficiently. It works by indexing vast amounts of data from web pages and other online resources, organizing this information into a search engine, it database. When a user enters a query into a search engine, it processes the request using complex algorithms to retrieve the most relevant results from its index, displaying them in an ordered list on the search results page. Search engines are essential tools for navigating the immense and ever-growing content available on the web. The core components of a search engine include a web crawler, an indexer, and a search algorithm. The web crawler, also known as a spider or bot, systematically browses the internet, visiting websites and



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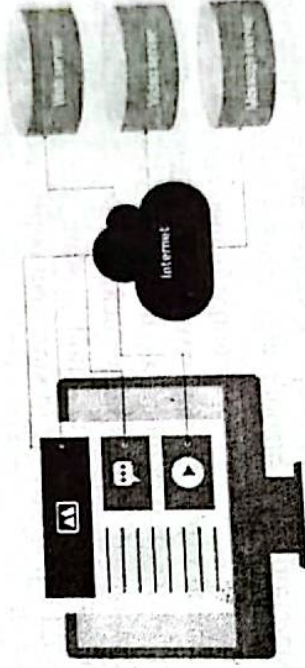
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collecting data from their pages. This data is then processed by the indexer, which categorizes and stores it in a structured manner, making it easier to retrieve during searches. The search algorithm, often proprietary and highly sophisticated, determines how to rank the results based on various factors such as keyword relevance, page quality, user location, and recency of the content. This ensures that users receive the most pertinent information in response to their queries.

In addition to basic web search, many search engines offer specialized search services like image search, video search, news search, and map search, among others. They also provide features such as autocomplete suggestions, which help users refine their queries, and personalized search results based on past search behavior. Popular search engines like Google, Bing, and Yahoo! have become integral to daily internet use, guiding users to find everything from basic information and academic research to shopping options and entertainment. With advancements in artificial intelligence and machine learning, search engines are continuously improving their ability to understand user intent and provide more accurate, context-aware results.

8.3 HTTP (Hypertext Transfer Protocol)

HTTP (Hypertext Transfer Protocol) is the foundational protocol used for transmitting data on the World Wide Web. It defines the rules and standards that web browsers, servers, and other web clients must follow to communicate and exchange information over the internet. HTTP operates as a request-response protocol in the client-server model, where a client, typically a web browser, sends a request to a web server, and the server responds by delivering the requested resources, such as HTML documents, images, or other content. HTTP is a stateless protocol, meaning that each request from a client to a server is treated as an independent transaction with no connection to any previous requests. This design allows HTTP to be simple and efficient, but it also means that additional mechanisms, like cookies or sessions, are required to maintain context or state between different interactions, such as keeping a user logged in across multiple pages of a website.



8.1 How HTTP Works

The communication process in HTTP typically involves the client sending an HTTP request message, which includes methods like GET (to retrieve data), POST (to submit data to be processed), PUT (to update data), and DELETE (to remove data). The server then processes this request and returns an HTTP response message containing the requested resource or a status code indicating the outcome of the request (e.g., 200 for success, 404 for not found, 500 for server error).

Over time, HTTP has evolved to improve security and performance. The most significant development in this regard is HTTPS (Hypertext Transfer Protocol Secure), which adds a layer of encryption using SSL/TLS protocols. HTTPS ensures that data transmitted between the client and server is secure, protecting it from interception or tampering. This security enhancement is particularly important for sensitive transactions, such as online banking, shopping, or any activity involving multiple requests and responses to be multiplexed over a single connection, reducing latency and enhancing the overall user experience on the web.

8.4 HTTPS

HTTPS (Hypertext Transfer Protocol Secure) is an enhanced version of HTTP designed to provide secure communication over the internet. It combines the standard HTTP protocol with SSL/TLS (Secure Sockets Layer/Transport Layer Security) encryption, ensuring that data transmitted between a user's web browser (or client) and a web server is protected from interception, tampering, or eavesdropping by malicious parties. HTTPS is crucial for safeguarding sensitive information such as passwords, credit card numbers, and personal data during online transactions.



https://

The key difference between HTTP and HTTPS lies in the security layer added by SSL/TLS. When a user connects to a website using HTTPS, the web server and the browser establish a secure connection through a process called an SSL/TLS handshake. During this handshake, the server presents a digital certificate to the browser, which verifies the authenticity of the server. Once the identity is confirmed, both the client and server agree on an encryption method and generate session keys for secure communication. This encryption ensures that even if data is intercepted, it cannot be read or altered without the correct decryption key.

One of the most visible indicators of HTTPS is the presence of a padlock icon in the browser's address bar, along with the "https://" prefix in the URL. In some cases, HTTPS-enabled websites also display a green address bar or an organization's name, particularly if they use an Extended Validation (EV) SSL certificate, which undergoes more rigorous verification. HTTPS has become the standard for all websites, not just those handling sensitive data, as it also protects against various cyber threats like man-in-the-middle attacks and data injection.

Moreover, search engines like Google give preference to HTTPS sites in their search rankings, encouraging website owners to adopt this protocol. The widespread use of HTTPS is part of a broader push towards creating a more secure and privacy-respecting internet, ensuring that users can trust the integrity and confidentiality of their online interactions.

8.5 HTML

HTML (Hypertext Markup Language) is the standard markup language used to create and structure content on the World Wide Web. It defines the structure and layout of a web page by using a series of elements and tags, which are interpreted by web browsers to render the content visually on the user's screen. HTML provides the framework for organizing text, images, links, forms, and other multimedia elements, allowing developers to create interactive and visually appealing web pages.



HTML documents are composed of a series of elements, each represented by a tag enclosed in angle brackets, such as `<html>`, `<head>`, `<body>`, `<p>`, and `<div>`. These tags often come in pairs, with an opening tag (e.g., `<p>`) and a closing tag (e.g., `</p>`) that surrounds the content. The structure of an HTML document typically begins with the `<!DOCTYPE html>` declaration, which defines the document type and version of HTML being used, followed by the `<html>` tag that encompasses the entire content of the page. Inside the `<html>` tag, the `<head>` section contains metadata, such as the page title and links to stylesheets or scripts, while the `<body>` section contains the actual content that will be displayed to the user.

HTML elements can be nested within each other, allowing for complex layouts and structures. For example, a `<div>` tag can contain multiple `<p>` (paragraph) tags, each of which can include other inline elements like `<a>` (links) or `<img>` (images). HTML also supports attributes, which provide additional

8.7 JavaScript

JavaScript is a versatile, high-level programming language that is widely used to create interactive and dynamic content on the web. Unlike HTML and CSS, which are primarily concerned with the structure and style of web pages, JavaScript adds functionality, allowing developers to implement complex features such as animations, form validation, content updates without page reloads (AJAX), and much more. It is an essential component of the web development trifecta, alongside HTML and CSS, enabling developers to build rich, interactive user experiences.

JavaScript is executed on the client side, meaning it runs directly in the user's web browser rather than on the server. This allows for immediate interaction and responsiveness, as the browser can process JavaScript code without needing to communicate with the server for every action. For example, JavaScript can be used to create interactive elements like drop-down menus, sliders, and modal windows, or to validate form inputs before they are submitted to the server, enhancing user experience and reducing server load.

One of the most powerful features of JavaScript is its ability to manipulate the Document Object Model (DOM), which represents the structure of a web page. By accessing and modifying the DOM, JavaScript can dynamically change the content, structure, and style of a web page after it has been loaded. For instance, JavaScript can be used to add new elements to a page, update the text or images based on user input, or change the style of an element when a user interacts with it (e.g., hovering over a button). JavaScript is also the foundation for many modern web development frameworks and libraries, such as React, Angular, and Vue.js, which help developers build complex, single-page applications (SPAs) that offer a smooth, app-like experience within a web browser. These tools leverage JavaScript to manage the state of an application, handle user input, and update the user interface in real-time.

In addition to its use in front-end development, JavaScript has also become a significant player on the server side with the advent of Node.js. Node.js allows developers to use JavaScript for server-side scripting, enabling them to build scalable, high-performance web applications with a unified programming language across both client and server.

JavaScript's flexibility, ease of use, and vast ecosystem of tools and libraries have made it one of the most popular programming languages in the world. It continues to evolve, with new features and improvements being added regularly, ensuring that it remains at the forefront of web development and beyond.

8.8 Web Server

A web server is a specialized software or hardware system that hosts websites and delivers web content to users over the internet. It plays a crucial role in the client-server architecture of the web, where it responds to requests from clients, typically web browsers, by serving up the requested web pages, images, videos, or other resources. The web server's primary function is to handle incoming HTTP or HTTPS requests and provide the appropriate responses, enabling users to access websites and web applications.

